INCH-POUND
MIL-DTL-24643/58G
27 November 2023
SUPERSEDING
MIL-DTL-24643/58F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LS2CS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, tin-coated, 26 AWG with 7 strands, ASTM B286.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation nominal wall thickness 0.015 inch; wall thickness may vary from nominal as necessary in order for completed cable to meet the specified electrical requirements. Twisted pair identification code shall be applied by method 6.
Fourth	-	Two conductors (one black and one white) cabled together to form a pair in accordance with table I. All pairs to be left-hand lay.
Fifth	-	The specified number of pairs (see table II) cabled together with a lay in accordance with table III. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Braided shield of tin-coated copper.
Eighth	-	Spirally-wrapped 0.002-inch thick polyethylene terephthalate (polyester film) separator tape, with 25 percent minimum overlap.
Ninth	-	Braided shield of tin-coated copper.
Tenth	-	An optional binder.
Eleventh	-	Cross-linked polyolefin jacket. See table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Pair lay.

LS2CS-77	LS2CS-60	LS2CS-42	LS2CS-18	LS2CS-6	65	66	67	62, 68, 71, 74, 77	63, 69, 72, 75	61, 64, 70, 73, 76	Pair number
					53	54	55	45, 48, 51, 56, 59	43, 46, 49, 52, 57, 60	44, 47, 50, 58	
					27, 39	28, 40	29, 41	19, 22, 25, 30, 33, 36, 42	20, 23, 26, 31, 34, 37	21, 24, 32, 35, 38	
					13	14	15	7, 10, 16	8, 11, 17	9, 12, 18	
					1	2	3	4	5	6	
Cable type and size					1.6	3.5	2.5	4.5	2.1	2.9	
					Length of lay (inches)						

TABLE II. Cable details.

Military part no. M24643/58	Type and size	Number of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UD	LS2CS-6	6	0.063	0.400	0.430
-02UD	LS2CS-18	18	0.063	0.550	0.590
-03UD	LS2CS-42	42	0.063	0.750	0.800
-04UD	LS2CS-60	60	0.082	0.880	0.930
-05UD	LS2CS-77	77	0.085	1.000	1.070

TABLE III. Cable lay.

	LS2CS-6		LS2CS-18		LS2CS-42		LS2CS-60		LS2CS-77	
	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)
Core	6	3.5	6	3.5	1	1.6	6	3.5	3	2.75
Layer 1	--	--	12	6.0	7	4.0	12	6.5	9	5.0
Layer 2	--	--	--	--	14	6.5	18	8.0	15	7.5
Layer 3	--	--	--	--	20	8.5	24	10.0	22	10.0
Layer 4	--	--	--	--	--	--	--	--	28	12.0

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	45.19
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to overall shield	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to overall shield	500
Overall shield to overall shield	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Characteristic impedance, at 1 megahertz (ohms)	135±15

Group B:

Cold bending (cable)	No failure
Bending endurance at 77±3.6 °F (25±2 °C) (cycles, min.)	25
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

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Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50

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Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2CS (all sizes)	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-074)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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